- *Discovery*—The basis for Cluster Bootstrap and many other Akka modules. It provides a lookup service for static configurations, as well as dynamic DNS, plus a specific implementation for Kubernetes, Amazon Web Services (AWS), Consul, or Marathon.

It is useful to have a mental model of these two major categories: `akka-cluster` with its modules and the `akka-management` core with its extensions. In this chapter, you learn about using `akka-cluster` to form a cluster and `akka-management` and `akka-management-cluster-http` to work on the nodes of the cluster over HTTP. The next section covers how a node becomes a member of the cluster, how to listen for membership events, and how to detect when nodes in the cluster have crashed.

NOTE The source code for this chapter is available at www.manning.com/books/akka-in-action-second-edition or https://github.com/franciscolopez sancho/akka-topics/tree/main/chapter08a and https://github.com/franciscolopezsancho/akka-topics/tree/main/chapter08b. You can find the contents of any snippet or listing in the .scala file with the same name as the class, object, or trait. The code in this chapter is grouped into two projects in the Git repository with different executables and configurations. This separation helps you run them separately without much extra configuration and lets us map ideas from the chapter text to the projects. The simplest cluster application is in chapter08a, and a more complex example of a cluster with word-counting logic is in chapter08b. This grouping of projects based on executables and ideas will continue throughout the book.